

Measurement

Perimeter & Area · Composite Figures · Surface Area · Volume · Pyramids, Cones & Spheres · Optimization

MTH1W · Homework — Independent Practice

Name: _____

Date: _____

Goal: Practise perimeter, area, surface area, volume and optimization independently.

Show your full working in the space provided under each question. Use $\pi \approx 3.14$ and round π -answers to **1 d.p.** Levels: L1 fluency, L2 application, L3 thinking. Worked solutions are in the Answer Key.

Level 1 — Fluency

1. Find the area of a rectangle 9 cm long and 4 cm wide. **[Rectangle area]**

2. Find the perimeter of a square with side length 8 cm. **[Square perimeter]**

3. Find the area of a triangle with base 12 cm and height 5 cm. **[Triangle area]**

4. A circle has radius 7 cm. Find its circumference ($\pi \approx 3.14$, 1 d.p.). **[Circle circumference]**

5. Find the volume of a cube with edge length 4 cm. **[Cube volume]**
6. A cylinder has radius 2 cm and height 9 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.).
[Cylinder volume]

Level 2 — Application

7. A parallelogram has base 15 cm and height 6 cm. Find its area. **[Parallelogram area]**
8. A trapezoid has parallel sides 10 cm and 6 cm and a height of 5 cm. Find its area.
[Trapezoid area]

9. A rectangle 12 cm by 5 cm has a semicircle of diameter 5 cm on one end. Find the total area ($\pi \approx 3.14$, 1 d.p.). **[Composite area]**
10. A rectangular prism is 5 cm by 3 cm by 2 cm. Find its surface area. **[Prism surface area]**
11. A cylinder has radius 4 cm and height 8 cm. Find its surface area ($\pi \approx 3.14$, 1 d.p.). **[Cylinder surface area]**
12. A cone has radius 3 cm and height 7 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.). **[Cone volume]**

Level 3 — Thinking

13. A sphere has radius 3 cm. Find its volume ($\pi \approx 3.14$, 1 d.p.). **[Sphere volume]**
14. A sphere has radius 5 cm. Find its surface area ($\pi \approx 3.14$, 1 d.p.). **[Sphere surface area]**
15. A square-based pyramid has base edge 8 cm and height 9 cm. Find its volume. **[Pyramid volume]**

16. A cylinder of radius 4 cm and height 10 cm has a cylindrical hole of radius 2 cm drilled straight through it. Find the remaining volume ($\pi \approx 3.14$, 1 d.p.). **[Volume with hole]**
17. A cone has radius 5 cm and slant height 13 cm. Find its surface area ($\pi \approx 3.14$, 1 d.p.). **[Cone surface area]**
18. A farmer has 100 m of fencing for a rectangular field and wants the maximum area. Find the dimensions and the maximum area. **[Optimization]**